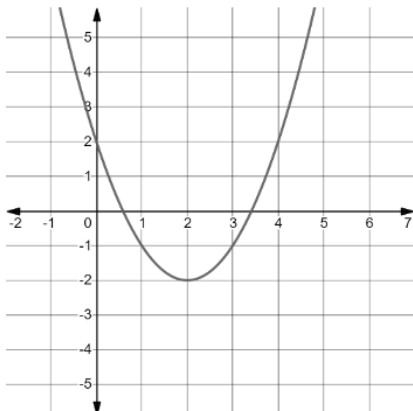


Algebra 1
Unit 13
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

The table shows three different functions.

Function A	Function B	Function C												
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>3</td></tr><tr><td>-1</td><td>6</td></tr><tr><td>0</td><td>9</td></tr><tr><td>1</td><td>12</td></tr><tr><td>2</td><td>15</td></tr></table>	x	y	-2	3	-1	6	0	9	1	12	2	15		$y = 5(1.4)^x$
x	y													
-2	3													
-1	6													
0	9													
1	12													
2	15													

1. Determine which type of function that each represents.

Function A	Function B	Function C
☐ Exponential ☒ Linear ☐ Quadratic	☐ Exponential ☐ Linear ☒ Quadratic	☒ Exponential ☐ Linear ☐ Quadratic

2. Which function has the largest y -intercept?

Function A

3. Compare the ranges of the functions and indicate any similarities and differences.

Answers may vary.

$A \rightarrow -\infty < y < \infty$

$B \rightarrow y \geq -2$

$C \rightarrow y > 0$

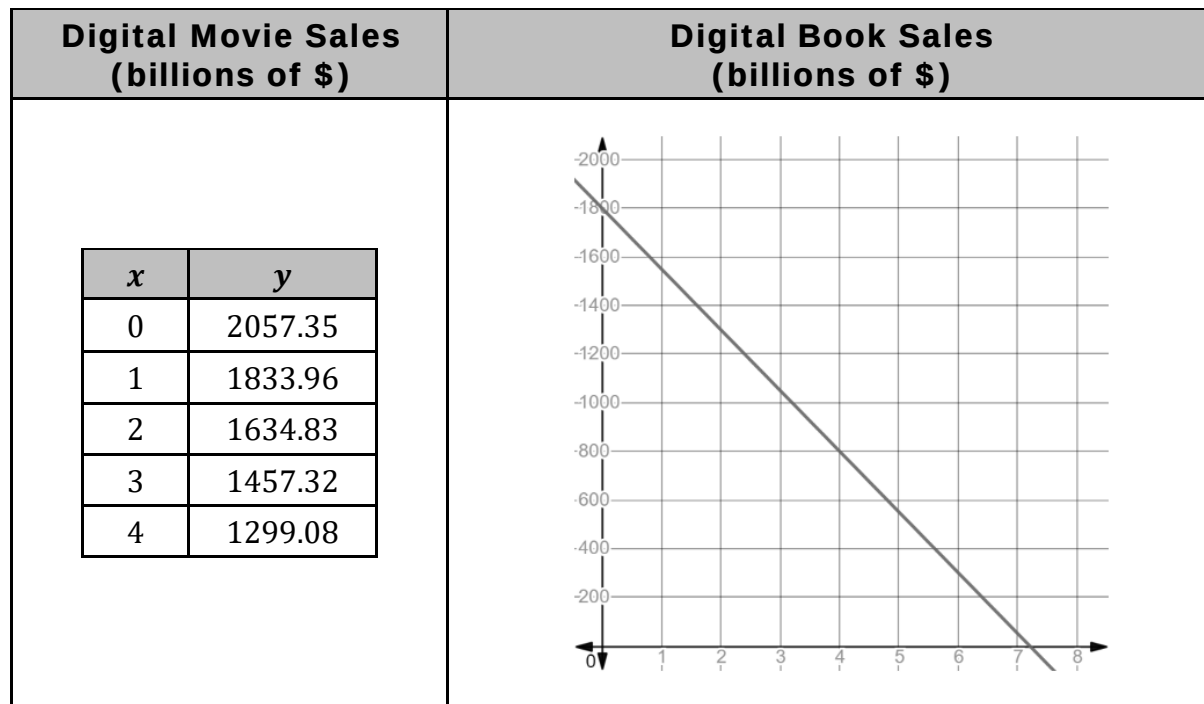
4. Compare the domains of the functions and indicate any similarities and differences.

All have a domain of all real numbers.

5. As $x \rightarrow \infty$, which function has larger values of y ?

Function C

The table shows a table of values representing the revenue, in billions of dollars, from sales of digital movies, where $x = 0$ is 2010, and a graph representing the revenue, in billions of dollars, from sales of digital books, where $x = 0$ is 2010.



6. Approximate the x -intercept for the function that represents the sale of digital books.

 ≈ 7.3

7. Interpret the x -intercept of the function that represents the sale of digital books.

Digital book sales would reach 0 in 2017.

8. What is the domain of the function representing the sale of digital books?

0 to approximately 7.3
 $0 \leq x \leq 7.3$

9. Which function has a larger y -intercept?

Movie sales

10. What type of function best describes the table of values that represents digital movie sales? Explain your choice. Explanations may vary.

- Exponential
 - Linear
 - Quadratic